

PROGRAM 10

MONITORING KUBERNETES CLUSTER USING DASHBOARD AND CLI

PROJECT STRUCTURE:

```
prog10/
|
|— deployment.yaml
|— service.yaml
```

STEP 1: Start Minikube

```
minikube start --cpus=2 --memory=4096
```

STEP 2: Launch Kubernetes Dashboard

```
minikube dashboard
```

STEP 3: Create deployment.yaml

STEP 4: Create service.yaml

```
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog10$ cat deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog10$ cat service.yaml
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: NodePort
```

STEP 5: Apply deployment and service

```
kubectl apply -f deployment.yaml
```

```
kubectl apply -f service.yaml
```

STEP 6: Check pods

```
kubectl get pods
```

STEP 7: Check nodes

```
kubectl get nodes
```

STEP 8: Check services

```
kubectl get svc
```

STEP 9: View pod logs

```
kubectl logs nginx-deployment-6f9664446b-bml6q
```

STEP 10: Describe pod

```
kubectl describe pod <pod-name>
```

STEP 11: Access application

```
minikube service nginx-service
```

STEP 12: Monitor using Dashboard

Observe pods, deployments, and services in the Kubernetes Dashboard opened earlier.

STEP 13: Clean up resources

```
kubectl delete deployment nginx-deployment
```

```
kubectl delete service nginx-service
```

OUTPUT:

Kubernetes cluster resources were monitored using both CLI and Dashboard. The application was successfully accessed through Minikube.

```

1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~$ cd prog10
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog10$ kubectl apply -f deployment.yaml
error: the path "deployment.yaml" does not exist
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog10$ ls
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog10$ nano deployment.yaml
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog10$ nano service.yaml
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment configured
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog10$ kubectl apply -f service.yaml
service/nginx-service created
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
kubia                               1/1     Running   3 (5m41s ago)    10d
nginx-deployment-6f9664446b-bml6q   1/1     Running   2 (5m51s ago)    6d23h
nginx-deployment-6f9664446b-rcf5d   1/1     Running   2 (5m51s ago)    6d23h
program-8                           1/1     Running   2 (5m41s ago)    6d23h
test-6bb654b8f8-p82wd               1/1     Running   3 (5m51s ago)    10d
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog10$ kubectl get nodes
NAME      STATUS   ROLES    AGE   VERSION
minikube  Ready    control-plane  23d   v1.34.0
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog10$ kubectl get svc
NAME                TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)          AGE
kubernetes          ClusterIP   10.96.0.1       <none>        443/TCP          23d
kubia               NodePort    10.100.58.146   <none>        8080:31766/TCP   10d
kubia-http          LoadBalancer 10.108.190.36   <pending>     8080:30716/TCP   10d
nginx-deployment    NodePort    10.109.18.4     <none>        80:30109/TCP     6d23h
nginx-service       NodePort    10.110.172.110 <none>        80:30965/TCP     11s
test                ClusterIP   10.104.246.143 <none>        80/TCP           17d
1RV24MC025_CHANDAN_K@chandan-Vivobook-ASUSLaptop-M6500QC-M6500QC:~/prog10$ kubectl logs nginx-deployment-6f9664446b-bml6q
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration

```

The screenshot displays the Kubernetes dashboard interface. The left sidebar contains a navigation menu with categories like Workloads, Service, Config and Storage, Cluster, and Network Policies. The main content area is titled 'Workload Status' and features three large green circular progress indicators for Deployments, Pods, and Replica Sets, each labeled 'Running: 1'. Below this, the 'Deployments' section shows a table with one entry: 'nginx-deployment' using 'nginx:latest' image, with 2/2 pods and created 'a minute ago'. The 'Pods' section shows a table with two entries, both 'nginx-deployment-6f9664446b-...' using 'nginx:latest' image, with status 'Running', 0 restarts, and created 'a minute ago'.

Name	Images	Labels	Pods	Created
nginx-deployment	nginx:latest	app: nginx	2 / 2	a minute ago

Name	Images	Labels	Node	Status	Restarts	CPU Usage (cores)	Memory Usage (bytes)	Created
nginx-deployment-6f9664446b-26k2g	nginx:latest	app: nginx pod-template-hash: 6f9664446b	minikube	Running	0	-	-	a minute ago
nginx-deployment-6f9664446b-777vh	nginx:latest	app: nginx pod-template-hash: 6f9664446b	minikube	Running	0	-	-	a minute ago